Comparing Anomaly Detection Methods for Medicare Provider Fraud

STAT 790 Final Project Report
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Abstract

I compare anomaly detection and supervised classification for flagging potentially fraudulent Medicare providers. The data is the Kaggle Healthcare Provider Fraud Detection set: 558,211 inpatient and outpatient claims, beneficiary demographics, and provider-level fraud labels, which I aggregated into 40 provider-level features covering billing, diagnoses, physician networks, demographics, and temporal volatility. Four unsupervised models (Isolation Forest, LOF, an Autoencoder, and a VAE) go up against four supervised ones trained with SMOTE (Logistic Regression, Random Forest, XGBoost, a Neural Network), all evaluated on the same 30% held-out provider test set. The Autoencoder was the best unsupervised model (ROC-AUC 0.812, PR-AUC 0.332), narrowly ahead of Isolation Forest at 0.804. LOF collapsed at 0.432 — worse than random. Supervised models sat in a much higher band, with Logistic Regression at 0.953 and XGBoost giving the best precision/F1 tradeoff. On dollars, the Autoencoder's top-10% alert budget caught $68.8M of recoverable reimbursement on the test set (~$229M scaled to the full population); Logistic Regression reached $101.9M (~$340M scaled). The takeaway is sequential: deploy unsupervised on day one, then move to supervised as confirmed labels accumulate.

1. Introduction

1.1 The Medicare fraud problem

Medicare is huge and the payment rules are complicated, so even a low error rate ends up costing billions. CMS put the FY2025 fee-for-service improper payment rate at 6.55%, or $28.83 billion. GAO's number is bigger — over $100 billion in improper payments across Medicare and Medicaid in FY2023. Improper isn't the same as fraudulent: CMS is clear that the category sweeps in missing documentation, administrative errors, over- and underpayments, and cases reviewers couldn't resolve either way. But it sets the scale of the problem and explains why data-driven screening earns its keep. A fraud unit doesn't need a model to prove fraud. It needs a model to point at a short list of providers worth investigating, with as little wasted effort as possible.

1.2 The detection challenge

Several things make this hard. The data mixes dollar amounts, dates, diagnosis and procedure codes, physician IDs, demographics, and chronic-condition flags — the primary diagnosis field alone has 10,450 unique values, which kills any straightforward distance-based approach. Fraud is also rare at the provider level: 506 of 5,410 providers (9.4%) carry the PotentialFraud label. And in a real deployment, those labels may not exist yet. A payer often needs to rank suspicious

providers before investigators have adjudicated enough cases to train anything supervised. A payer may need to rank suspicious providers before investigators have adjudicated enough cases to train a supervised classifier. This is the cold-start setting that motivates unsupervised anomaly detection.

1.3 Project objectives and contributions

I extend the Slavov (2014) baseline in a few directions. The classical density-based approach gets replaced with Isolation Forest, an Autoencoder, and a VAE, though I kept LOF in as a sanity check against a classical baseline. Evaluation is on a real held-out test set rather than on the same data used to define suspiciousness — this matters more than it sounds, since the alternative quietly inflates everything. AUCs come with bootstrap confidence intervals so nothing gets read as a deterministic point estimate. And the model output gets translated into something operational: rank providers by suspicion, set a fixed alert budget, count recoverable reimbursement from true positives, charge investigation cost against false positives. The supervised and unsupervised models are then compared on the same test set, which is what makes it possible to put a number on what labels are worth.

2. Related Work and Baseline Paper

2.1 The Slavov (2014) baseline

The provided baseline paper posed healthcare fraud detection as a problem of anomaly detection among mixed claims data. Its main insight was that providers engaging in fraud may deviate from their own historic baseline, rather than simply being global outliers. It also introduced the practical complication of having mixed categorical and continuous features: diagnosis/procedure codes cannot be readily compared with ordinary Euclidean distance without some transformation. The baseline consequently drives this project's emphasis on provider-level aggregation and representation learning approaches. Its weaknesses form the basis for improvements in this work: limited use of held-out evaluation, reliance on classic methods that have known limitations in sparse mixed spaces, and lack of direct comparison between cold-start unsupervised fraud detection and steady-state supervised evasion.

2.2 Classical and modern anomaly detection

LOF asks whether a point sits in a sparse neighborhood relative to its neighbors. Intuitive, but it falls apart in high-dimensional tabular data — distances stop being informative and local density estimates get noisy. Isolation Forest sidesteps the problem entirely: random partitions of the feature space, and points scored by how few cuts it takes to isolate them. Autoencoders take a different angle, learning to reconstruct normal observations through a bottleneck and scoring fraud by reconstruction error. VAEs add a probabilistic latent layer with a KL penalty, which is great for generative work but, as it turned out here, over-regularizes a tabular dataset this size.

2.3 Supervised imbalanced classification

After labels have been created, classification of these labels is known as rare-event classification. The imbalance is extreme enough that using accuracy as a metric is deceiving: naively labeling every provider as legitimate would result in over 90% accuracy while finding zero fraud cases. SMOTE can help solve this issue by creating artificial minority-class observations by interpolation between neighboring observations of fraud. The supervised learning models used for this project range from simple and interpretable to complex and opaque: Logistic Regression (a transparent linear baseline), Random Forest (non-linear decision trees with built-in feature importances), XGBoost (sequentially-built gradient boosted trees which typically perform well on tabular datasets), and a Neural Network (a flexible non-linear classifier). ROC-AUC is used to score the quality of ranking of predictions for all possible thresholds, while PR-AUC is also used since the positive class is rare.

3. Data

3.1 Dataset overview and assembly

The data comes from the Kaggle Healthcare Provider Fraud Detection Analysis set: separate training files for provider labels, beneficiary records, inpatient claims, and outpatient claims. I only used the training files because they're the ones with PotentialFraud labels attached. I stacked the inpatient and outpatient claims into a single table with a ClaimType indicator, derived length of stay from the admission and discharge dates, joined beneficiary demographics by BeneID, and attached provider labels by Provider. That gave me 558,211 claims. The modeling unit, though, is the provider — fraud here is a billing pattern across many claims, not a single bad transaction.

Table 1. Dataset scale and modeling split

Object	Count / value	Interpretation
Provider labels	5,410 providers	Unit used for modeling
PotentialFraud providers	506 (9.4%)	Rare positive class at provider level
Beneficiary records	138,556	Patient demographics and chronic conditions
Inpatient claims	40,474	Hospital claims with length-of-stay fields
Outpatient claims	517,737	Ambulatory claims
Combined claims	558,211	Claim-level table after stacking
Train/test split	3,787 / 1,623 providers	Stratified 70/30 split; same seed in all notebooks
Test fraud count	152 providers	Real operational class balance preserved

Provider-level fraud is rarer than claim-level fraud because the fraudulent providers bill a lot more often and at higher amounts. That distinction matters. A claim-level model would end up dominated by high-volume providers, but the actual operational question is which providers to investigate. The EDA reimbursement comparison shows that the positive class isn't just more common in volume — it's much larger in dollars.

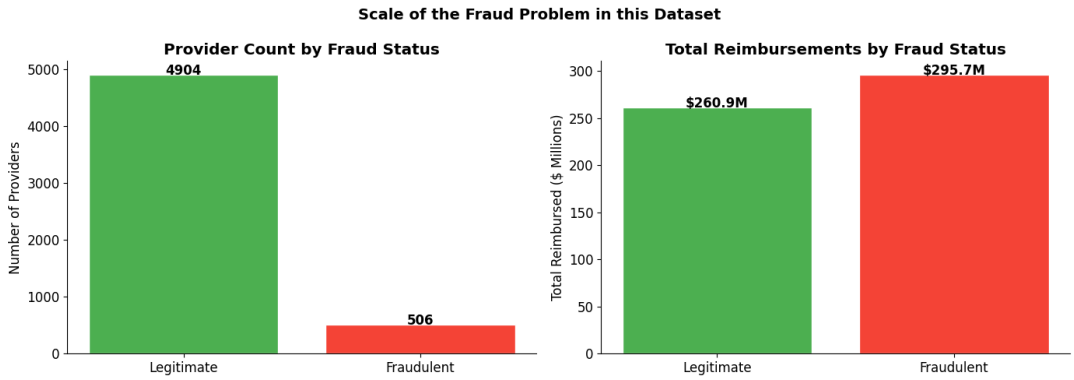

Figure 1. Provider label imbalance and average reimbursement by fraud label. Fraudulent providers are rare but financially much larger on average.

3.2 The mixed-data problem

The single biggest data issue is the mix of continuous billing variables and high-cardinality categorical codes. One-hot encoding every diagnosis and procedure field would blow the feature space up into something hopelessly sparse, and classical distance-based methods would break on it. That's why I aggregate at the provider level and turn the raw codes into behavioral summaries — unique diagnosis count, diagnosis entropy, diagnosis concentration, inpatient share, physician count, temporal volatility. The categorical mess gets compressed into interpretable provider behavior instead.

Table 2. Categorical cardinality in the claim-level data

Field	Unique values	Why it matters
ClmDiagnosisCode_1	10,450	Primary diagnosis alone creates a very sparse space
ClmDiagnosisCode_2	5,300	Secondary diagnosis adds high-dimensional sparsity
ClmDiagnosisCode_3 to _5	3,970-4,756	Additional codes reinforce mixed-data challenge
ClmProcedureCode_1	1,117	Procedure code variety is large but highly missing
ClmProcedureCode_2 / _3	300 / 154	Procedure fields are sparse and episodic
State / County	52 / 314	Geography can proxy practice mix and patient pool
Gender / Race / Renal / ClaimType	2-4	Low-cardinality demographic/claim-type variables

Missingness in this data isn't really missingness. The procedure-code fields are empty for most claims because most outpatient claims don't involve reportable procedures in those slots. Treating those gaps as noise would be wrong — they're a structural feature of how claims work. So the final provider features sidestep one-hot encoding of sparse raw codes and instead summarize diagnosis and procedure behavior through counts, concentration, entropy, and intensity measures.

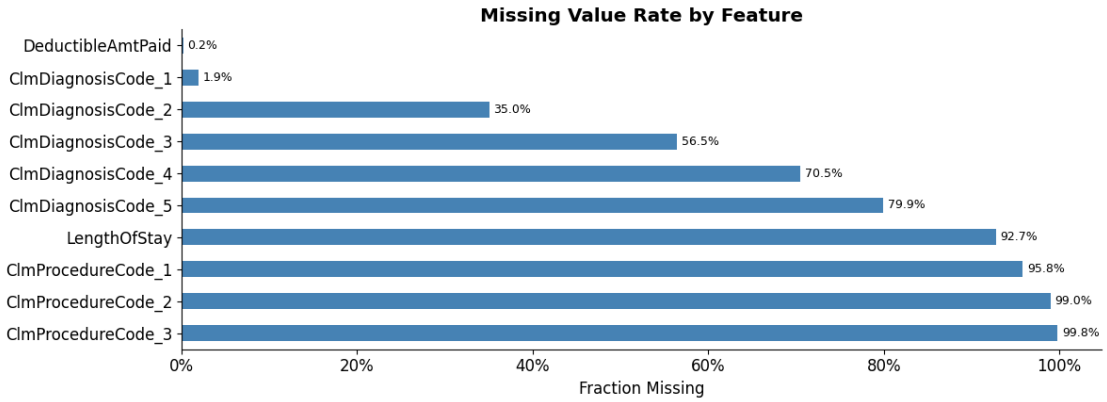

Figure 2. Missing-value rates for selected claim fields. Procedure-code missingness is structural and expected, not simply data quality failure.

3.3 Exploratory data analysis

EDA shows pretty clean separation between fraudulent and legitimate providers on volume and size. Fraudulent providers have more claims, more total reimbursement, more patients, and more diagnosis codes. The median legitimate provider had 27 claims and $15,055 in total reimbursement; the median fraudulent one had 155.5 claims and $373,450. Correlations with the fraud label come out highest for TotalReimbursed (0.576), UniqueDiagnosisCodes (0.400), UniquePatients (0.394), and TotalClaims (0.374). None of that proves fraud on its own, but it's enough signal to rank on.

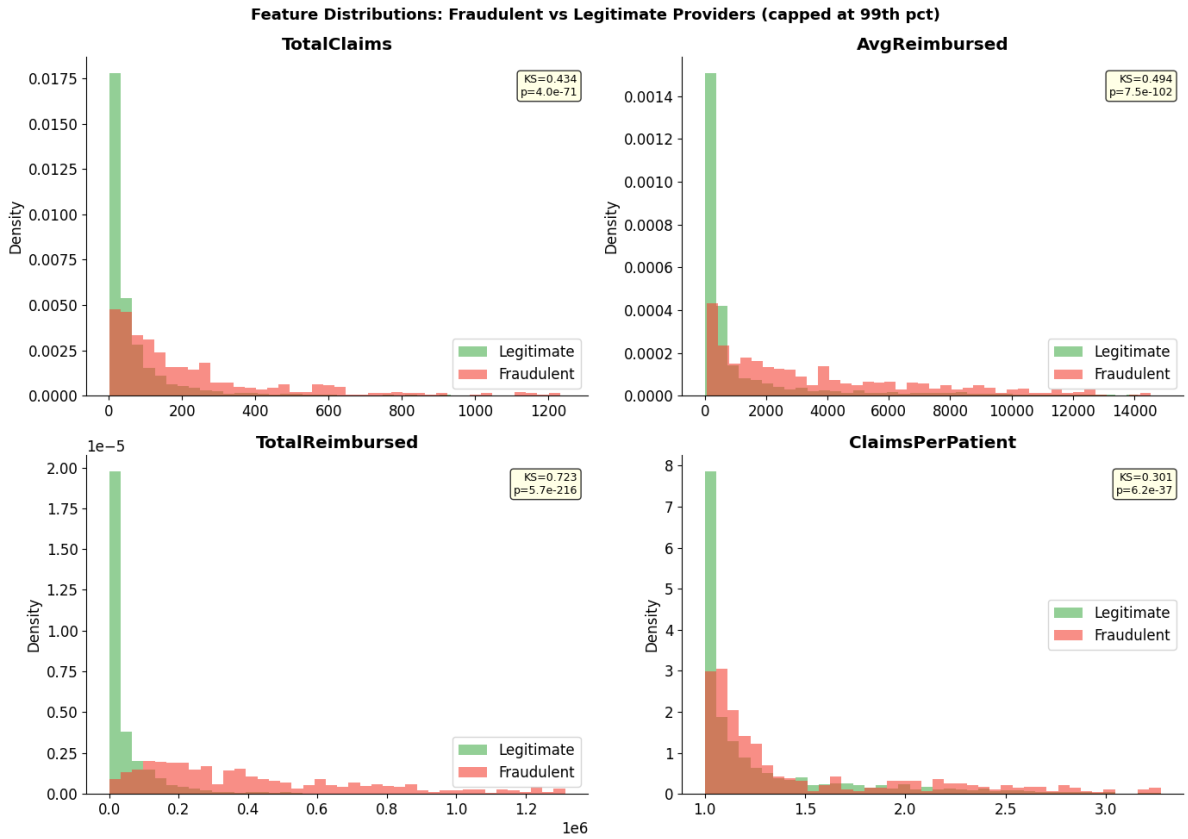

Figure 3. Selected provider-level distributions by fraud label. Fraudulent providers show visibly heavier tails in volume and reimbursement features.

I also replicated the baseline's temporal drift idea. The monthly average claim amount plot shows that provider trajectories over the year are all over the place. That supports keeping temporal volatility features in the model, but it's also a warning — billing changes can come from case mix, seasonality, or perfectly legitimate operational changes. Drift on its own isn't a fraud signature. The temporal features earn their place as incremental signal, not as a smoking gun.

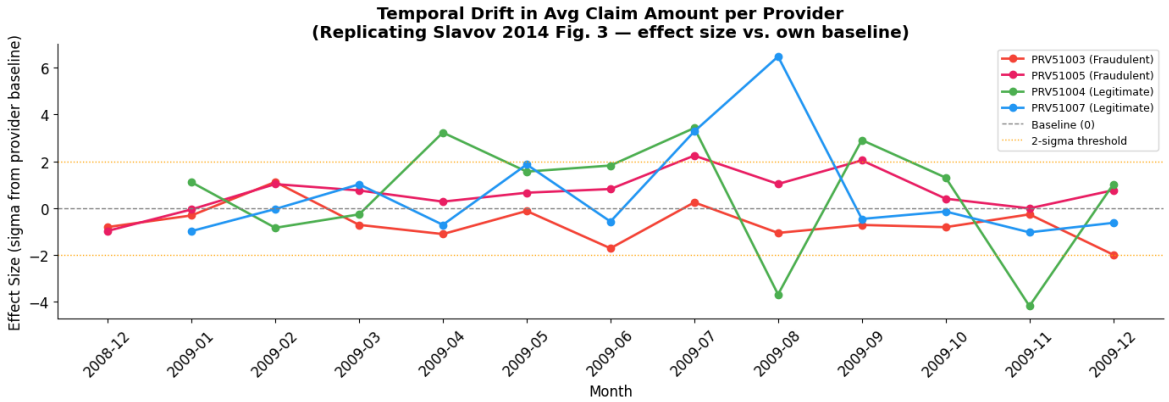

Figure 4. Temporal drift replication: monthly average claim amount by provider, illustrating movement away from provider-specific baselines.

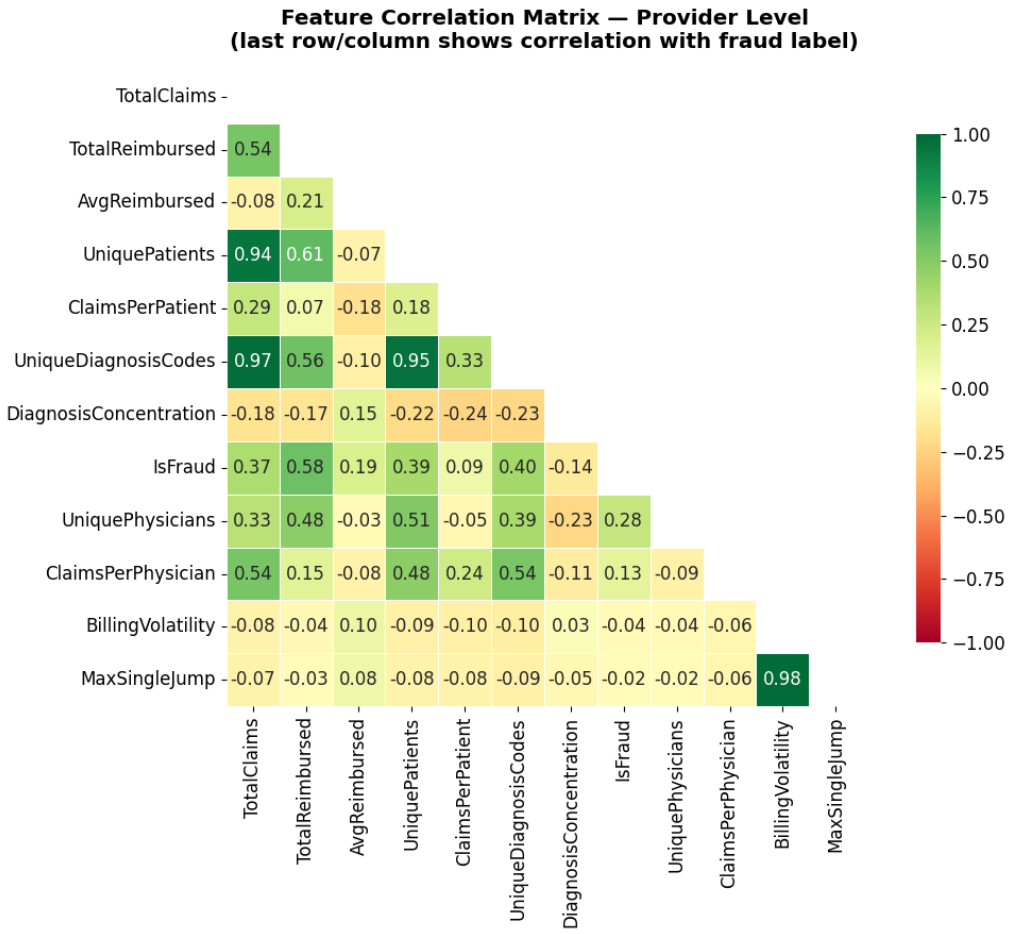

Figure 5. Provider-level feature correlation heatmap. Total reimbursement, diagnosis diversity, patient count, and claim volume are the strongest fraud-correlated signals.

Diagnosis and physician-network features add a second axis beyond raw volume. Diagnosis concentration by itself isn't a clean separator, but diagnosis diversity and physician counts both track suspicious providers consistently. That's why both groups end up in the 40-feature matrix.

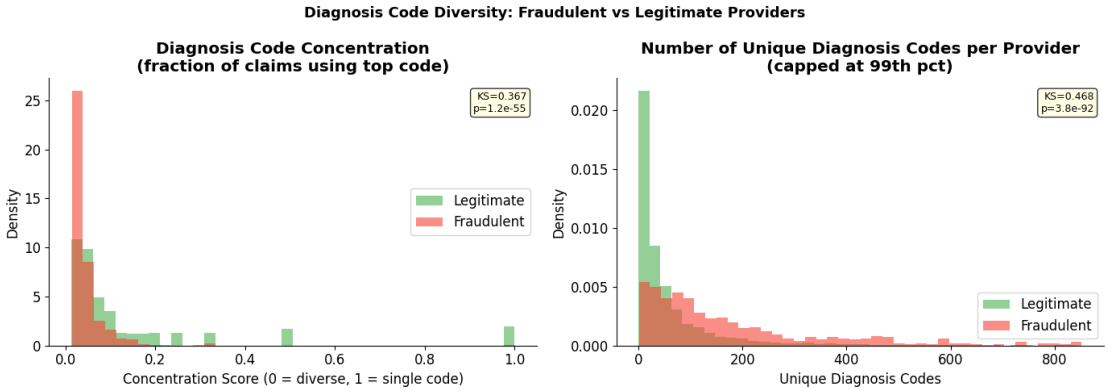

Figure 6. Diagnosis-code concentration and unique diagnosis codes by fraud label.

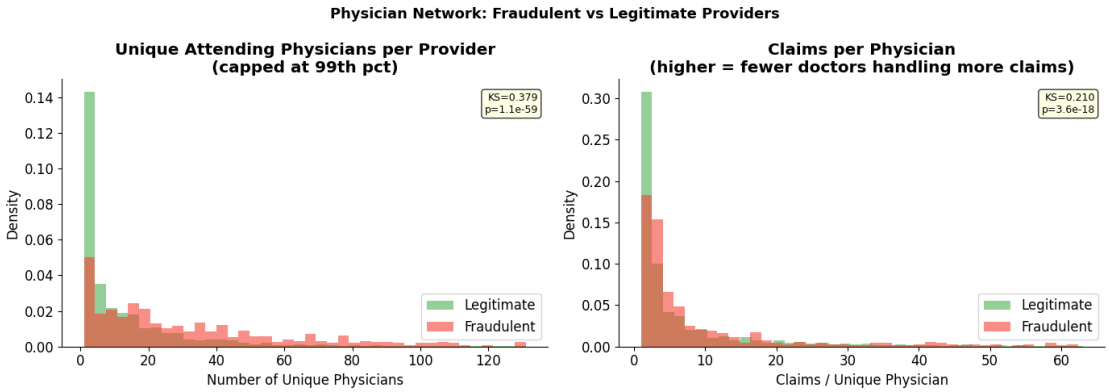

Figure 7. Physician-network features by fraud label.

4. Feature Engineering

4.1 Provider-level aggregation

Modeling happens at the provider level. The label is attached to providers, not claims, so the unit of prediction has to match. Aggregation also turns the messy high-cardinality code fields into something readable: one unusual claim might just be a complex patient, but unusual reimbursement patterns spread across hundreds of claims is something investigators can act on. As a side benefit, 558,211 claims collapse to 5,410 providers, which makes the bootstrap evaluation, model comparison, and financial analysis actually tractable.

4.2 The 40 features in five groups

I ended up with 40 numeric features, grouped by the kind of behavior they capture rather than by which table they came from. The grouping is the part that matters for an analyst: a provider can look off because of scale, diagnosis behavior, physician network shape, patient mix, or change over time — and those are different stories.

Table 3. Provider-level feature groups

Group	Representative features	Fraud intuition
Billing behavior	TotalClaims, TotalReimbursed, Avg/Median/Max/StdReimbursed, Avg/MaxDeductible, InpatientShare, length-of-stay statistics	Fraudulent providers may bill more often, bill larger amounts, or have unusual inpatient intensity
Diagnosis patterns	UniqueDiagnosisCodes, DiagnosisConcentration, DiagnosisEntropy	Suspicious providers may overuse certain codes or show atypical diagnosis variety
Physician network	UniquePhysicians, ClaimsPerPhysician	Provider-physician relationships may reveal concentration or unusually broad billing networks
Demographics and clinical burden	AvgAge, FemaleShare, RenalShare, chronic-condition means	Controls for patient mix so high spending is not automatically treated as fraud
Temporal volatility	MonthsActive, BillingVolatility, MaxSingleJump, ClaimsVolatility, TotalReimbursedVolatility	Sudden or unstable changes may indicate drift from a normal baseline

4.3 Missingness and scaling

Claims data has missingness baked in. Outpatient claims don't have inpatient length-of-stay fields. Most claims don't fill all the procedure-code slots. Some monthly volatility values are undefined when a provider barely operates across enough months. The pipeline replaces infinities with NaN, imputes with the training-set median, and then applies StandardScaler. For supervised models, the imputer and scaler are fit only on training and applied to test. For the Autoencoder and VAE, training uses only legitimate training providers, so the model learns what normal reconstruction looks like without ever seeing a fraud label. Both choices are about not leaking the test set.

5. Methodology

5.1 Train/test design

Both notebooks use the same random seed and the same stratified 70/30 provider split. The training set contains 3,787 providers, including 354 labeled fraud providers, while the held-out test set contains 1,623 providers, including 152 fraud providers. Stratification preserves the 9.4% fraud rate in both partitions. This design is central to the report because it allows a fair head-to-head comparison: supervised models, unsupervised models, ROC/PR curves, alert-budget thresholds, and ROI estimates all refer to the same test providers.

5.2 Unsupervised models

The unsupervised models are designed for the cold-start setting. Isolation Forest was fit on the full training set and evaluated with two thresholds: the theoretical Liu-Ting-Zhou anomaly threshold of score > 0.5 and fixed operational alert budgets at the top 5% and top 10% of suspicious providers. LOF was evaluated as the classical density-based baseline. The Autoencoder used a symmetric neural architecture with a compressed bottleneck and reconstruction error as the anomaly score. The VAE used a probabilistic latent layer and combined reconstruction loss with KL regularization. For the neural unsupervised models, fraud labels are not used in training; labels appear only when measuring test-set separation.

5.3 Supervised models and SMOTE

The supervised pipeline splits first, then SMOTE-balances only the training set. Pre-SMOTE that's 3,433 legitimate and 354 fraudulent providers (9.7:1). Post-SMOTE it's 3,433 of each, with 3,079 synthetic fraud examples. The test set stays untouched at its real 9.4% rate. Four models on top: L2 Logistic Regression, Random Forest with 300 trees, XGBoost with 500 boosting rounds, and a feed-forward neural net. That covers the spectrum from interpretable linear to flexible non-linear.

Order matters here. SMOTE before splitting would let synthetic fraud examples get generated from observations that later show up in test, which inflates everything. Splitting first means synthetic examples live only in training and the test set stays a realistic sample of real providers. That's especially important for the financial analysis — the ROI numbers depend on

real reimbursement amounts attached to real providers, and contaminating the test set would quietly break that.

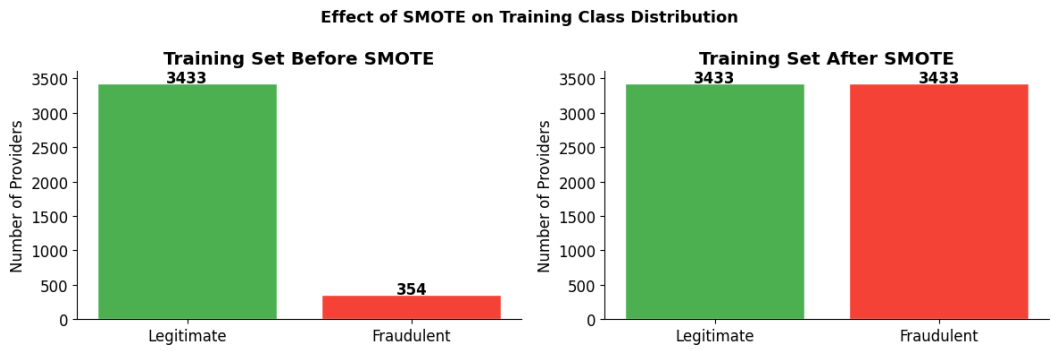

Figure 8. SMOTE balances the supervised training set only; the test set remains unchanged.

5.4 Evaluation and financial framework

ROC-AUC handles overall ranking quality; PR-AUC matters because the positive class is rare. Precision, recall, and F1 get reported at operational thresholds, and AUC estimates come with bootstrap confidence intervals from 500 test-set resamples. For the financial side, the review budget flags the top 10% of providers. Recoverable reimbursement for a given model is the total reimbursement paid to its true-positive flagged providers; investigation cost is the false-positive count times a per-investigation scenario of $500, $1,000, or $2,000. Net ROI is the difference. This is a prioritization upper bound, not a legal recovery guarantee — PotentialFraud labels aren't adjudicated convictions, and treating them as such would overstate things.

The calculation is Net ROI = (reimbursement to true-positive flagged providers) − (false positives × investigation cost). The test set is 30% of providers, so full-population estimates multiply by roughly 3.33. That scaling is for communicating magnitude only — the rigorous comparison is the head-to-head on the held-out test set.

6. Results - Unsupervised Models

6.1 Individual model performance

The cold-start finding is the centerpiece here. The Autoencoder comes out on top with ROC-AUC 0.812 and PR-AUC 0.332. Isolation Forest is right behind on ROC-AUC at 0.804, though it lags on PR-AUC. The VAE drops to 0.627. LOF lands at 0.432 — worse than random, with a confidence interval entirely below 0.5. At the top-10% budget, the Autoencoder flags 163 providers, catches 51 of the 152 fraudulent ones, and hits 0.313 precision. For a model that never saw a single confirmed fraud label, that's not bad.

Table 4. Unsupervised model performance on the held-out test set

Model / threshold	Flagged	Precision	Recall	F1	ROC-AUC (95% CI)	PR-AUC (95% CI)
Isolation Forest, theoretical s>0.5	130	0.238	0.204	0.220	0.804 [0.773, 0.831]	0.234 [0.190, 0.284]
Isolation Forest, top 10%	163	0.245	0.263	0.254	0.804 [0.773, 0.831]	0.234 [0.190, 0.284]
LOF, top 10%	163	0.067	0.072	0.070	0.432 [0.380, 0.476]	0.102 [0.074, 0.134]
Autoencoder, top 10%	163	0.313	0.336	0.324	0.812 [0.783, 0.839]	0.332 [0.265, 0.403]
VAE, top 10%	163	0.190	0.204	0.197	0.627 [0.582, 0.671]	0.203 [0.147, 0.260]

6.2 Score distributions and curves

The score plots match the table. Isolation Forest and the Autoencoder both push fraudulent providers toward higher anomaly scores, but the Autoencoder pulls the top tail apart more cleanly. LOF actually inverts the ranking — fraudulent providers often look *less* anomalous than legitimate ones, which is exactly what the sub-0.5 ROC-AUC says. ROC and PR curves tell the same story: Autoencoder and Isolation Forest dominate, VAE sits in the middle, and LOF isn't operationally usable here.

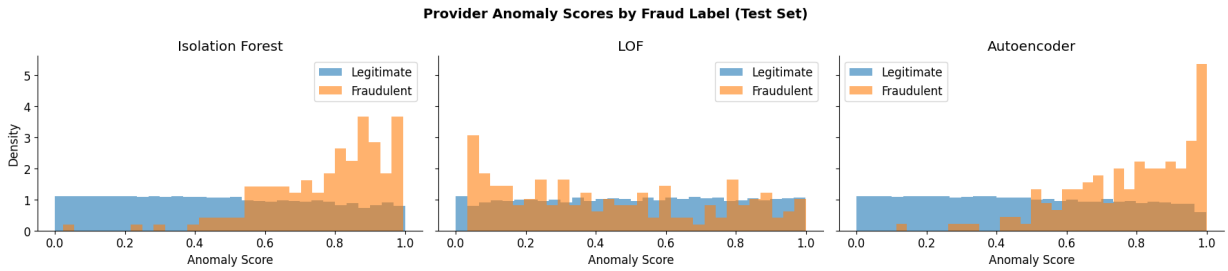

Figure 9. Unsupervised anomaly score distributions on the held-out test set.

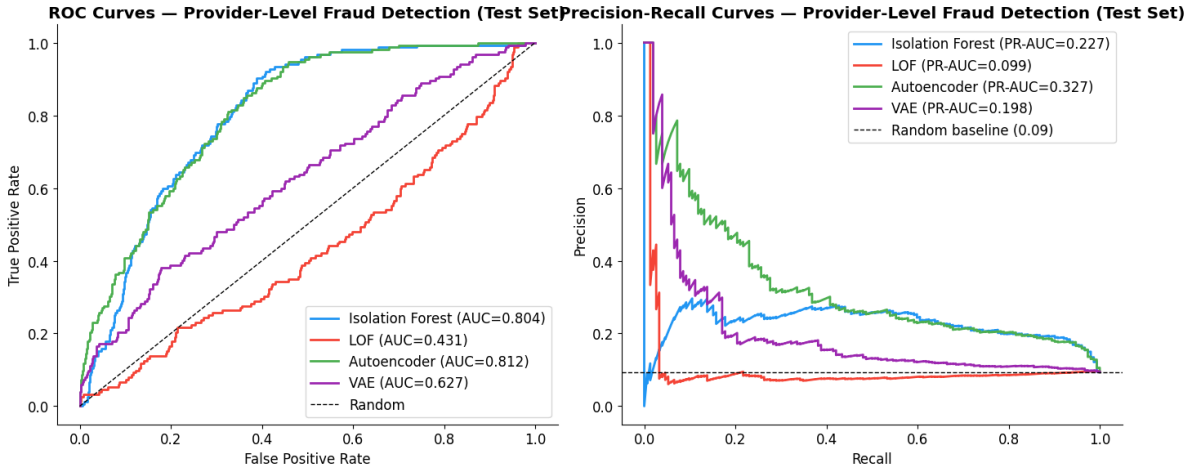

Figure 10. ROC and precision-recall curves for unsupervised models.

6.3 Alert-budget behavior

No fraud unit gets to investigate every provider, so the alert-budget sweep walks through what happens as the flagged fraction moves from 1% to 30%. Precision drops as the budget widens; recall climbs. The Autoencoder dominates precision across the low-budget region — the

region that actually matters — and stays the top F1 performer at both the 5% and 10% thresholds. Isolation Forest is the practical second choice. It's nearly tied on ROC-AUC and has a threshold rule that doesn't require knowing the true fraud rate, which is useful when you don't.

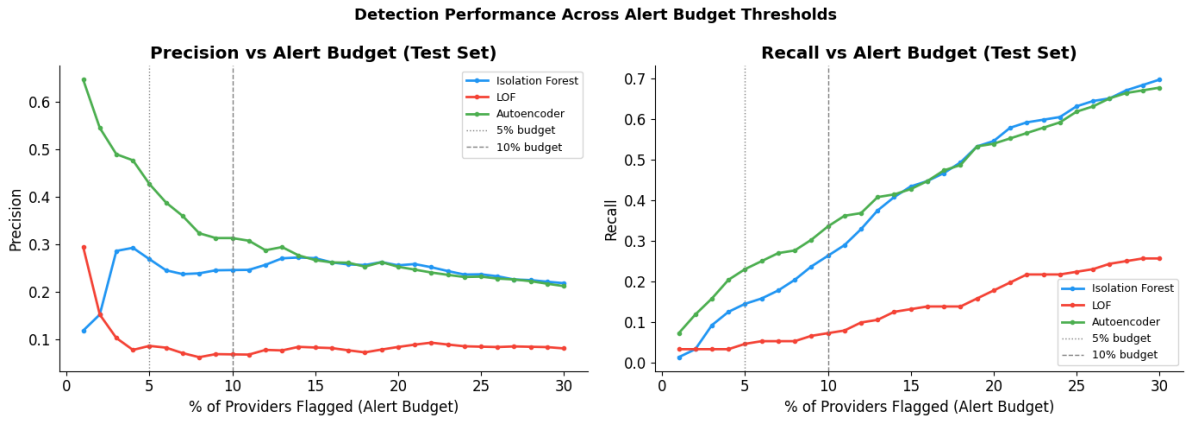

Figure 11. Precision and recall across investigation budgets for unsupervised models.

6.4 Why LOF fails and why AE beats VAE

The LOF result surprised me at first, but it makes sense in retrospect. LOF needs stable local neighborhoods to estimate density, and the 40-feature provider space doesn't provide them. Billing variables are heavily skewed, diagnosis-derived features are sparse, demographic controls are categorical-ish, and the temporal features are volatile. Nothing about that combination supports reliable density comparisons, so LOF ranks worse than random. The VAE underperforms the plain Autoencoder for a different reason: the KL term forces the latent space to be smooth and distributional, which is useful for generative modeling but appears to smooth away precisely the provider-level reconstruction failures that make fraud detectable. The simpler Autoencoder is better suited to this medium-sized tabular dataset.

The close Autoencoder-Isolation Forest result is also informative. Isolation Forest performs well because it does not try to estimate local density or fit a parametric distribution; it only asks whether a provider can be isolated quickly through random cuts. The Autoencoder performs slightly better because it can learn non-linear combinations of features that define normal provider behavior. For example, high total reimbursement may be normal for a provider with many patients and a high inpatient share, but unusual for a provider with few patients and concentrated diagnoses. Reconstruction error can capture those interactions more naturally than a univariate rule or a purely random partition.

7. Results - Supervised Models

7.1 Model performance

Once labels are in, the ceiling jumps. All four supervised models cluster between ROC-AUC 0.930 and 0.953. Logistic Regression ranks best by ROC-AUC and has the highest recall, catching 131 of the 152 fraud providers. XGBoost has the strongest F1 and the highest precision

— fewer legitimate providers flagged, but still 106 fraud providers caught. Random Forest sits between them, and the neural network is competitive but slightly lower on AUC. The fact that all four cluster so tightly says the engineered provider features are doing most of the work.

Table 5. Supervised model performance on the held-out test set

Model	Precision	Recall	F1	ROC-AUC (95% CI)	PR-AUC
Logistic Regression	0.478	0.862	0.615	0.953 [0.936, 0.968]	0.742
Random Forest	0.558	0.724	0.630	0.950 [0.930, 0.965]	0.679
XGBoost	0.639	0.697	0.667	0.947 [0.928, 0.964]	0.723
Neural Network	0.614	0.671	0.642	0.930 [0.908, 0.952]	0.689

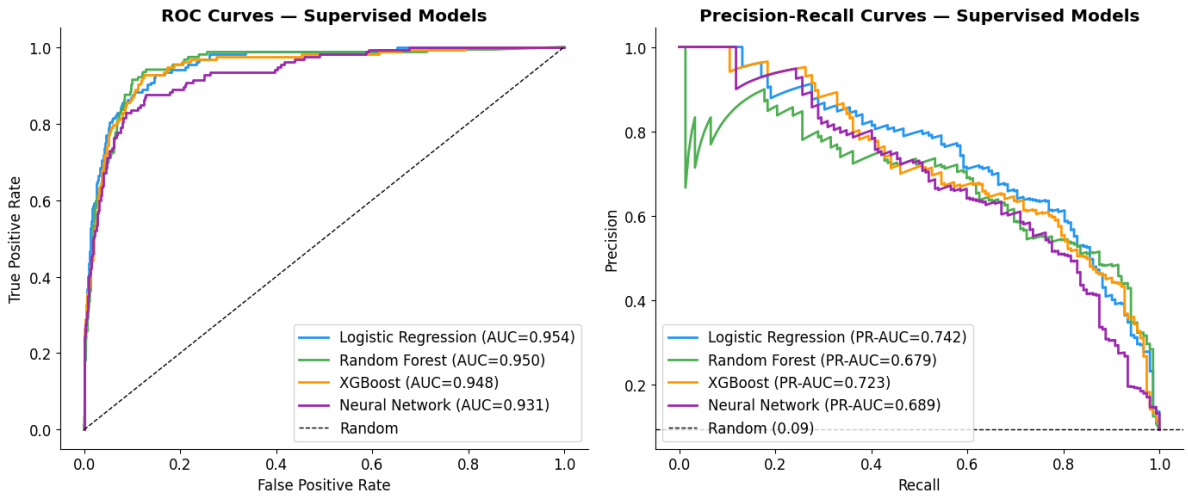

Figure 12. ROC and precision-recall curves for supervised models.

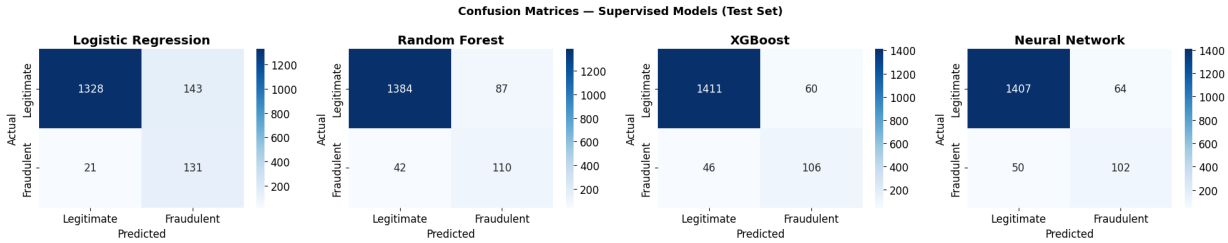

Figure 13. Supervised confusion matrices on the held-out test set.

7.2 Feature importances and operational tradeoffs

Analysts need to know why a provider got flagged, so interpretability is part of the deliverable. Logistic Regression coefficients give the cleanest read: higher TotalReimbursed, TotalClaims, UniqueDiagnosisCodes, UniquePatients, and the inpatient-related variables push providers toward fraud; some chronic-condition and demographic controls pull the other way. Random Forest and XGBoost emphasize non-linear split importance instead. XGBoost's top feature is TotalReimbursed by a wide margin, followed by MaxLengthOfStay, monthly volatility measures, UniqueDiagnosisCodes, and MaxDeductible. Reimbursement scale dominates, but diagnosis diversity and temporal structure are pulling their weight too.

Table 6. Interpretability summary across supervised models

Model	Most useful interpretation	Operational implication
Logistic Regression	Signed coefficients show which features increase or decrease fraud odds	Best for explaining broad direction of risk and communicating simple rules
Random Forest	Mean decrease in impurity identifies features that repeatedly split providers well	Best for robust non-linear feature screening
XGBoost	Boosted-tree importance emphasizes features that correct prior model errors	Best for high-confidence ranking under limited investigative capacity

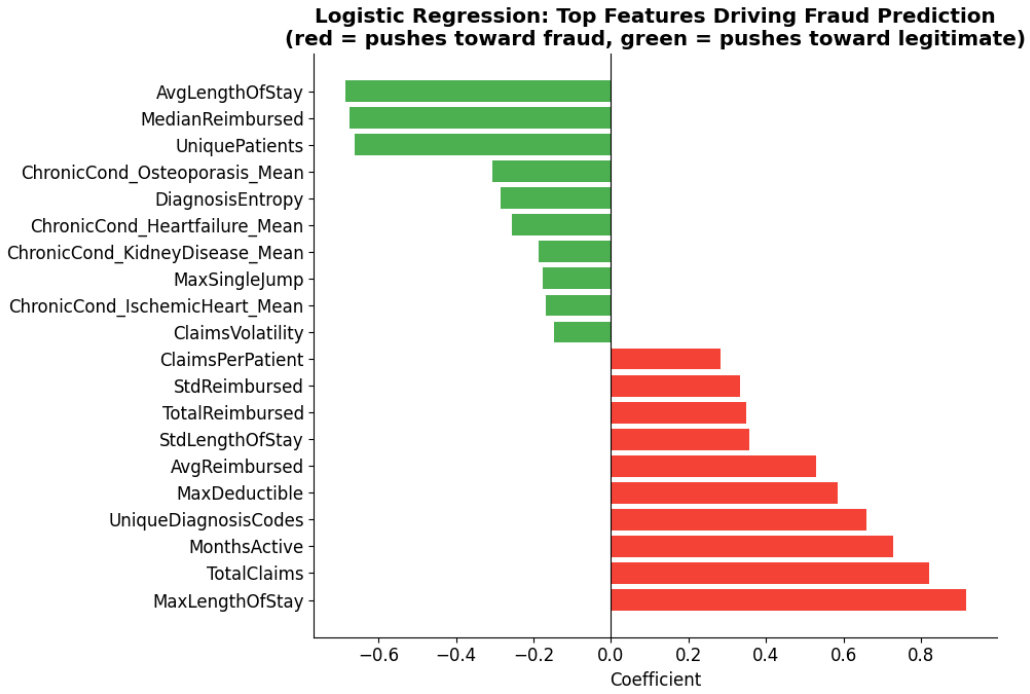

Figure 14. Logistic Regression coefficients: red features push toward fraud, green features push toward legitimate.

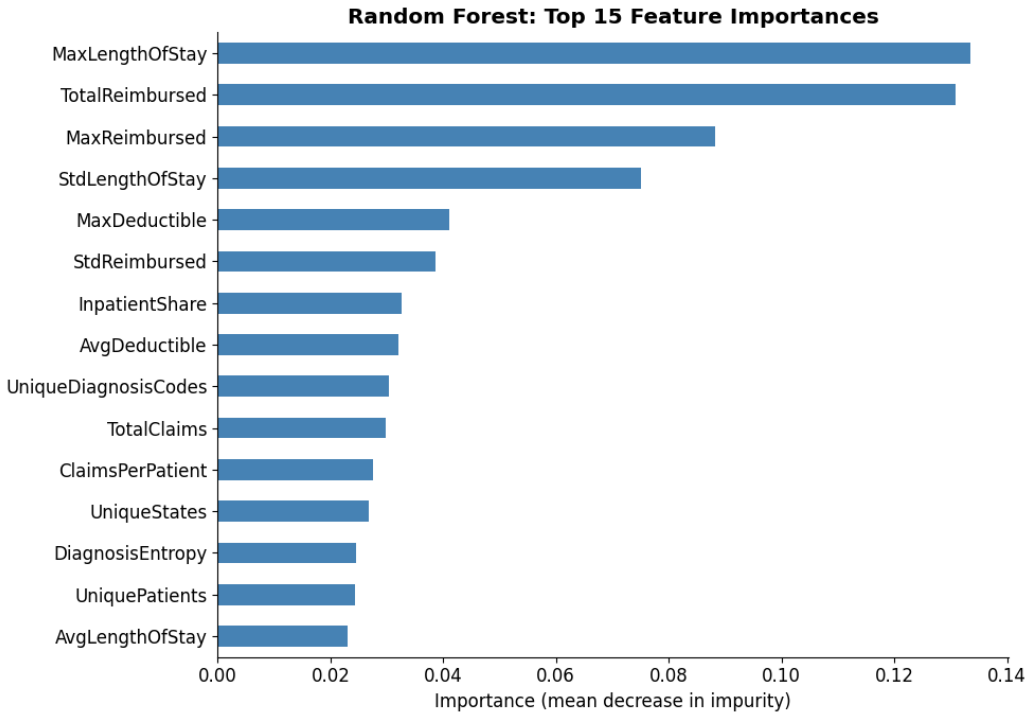

Figure 15. Random Forest feature importance from non-linear tree splits.

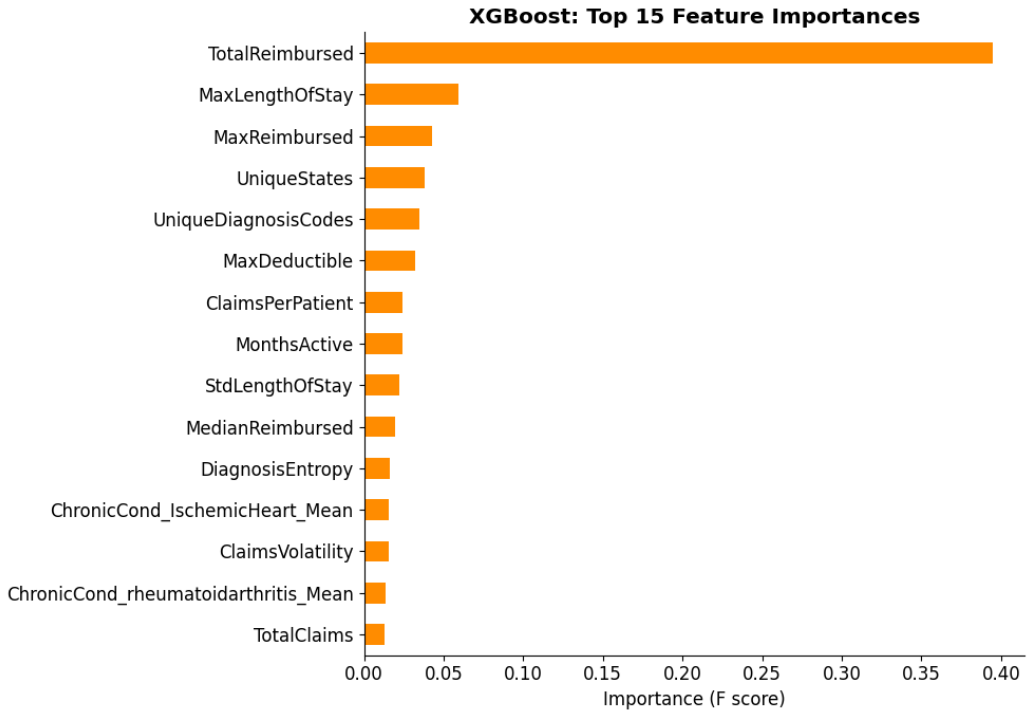

Figure 16. XGBoost feature importance. TotalReimbursed dominates, but length-of-stay, volatility, and diagnosis features also contribute.

Which model wins depends on what binds the fraud unit. If missed fraud is the costly failure mode, Logistic Regression is the pick — recall 0.862, precision 0.478. If investigator capacity is the binding constraint, XGBoost wins — precision 0.639, best F1, only 60 false positives at the default threshold. In practice a payer could run both: Logistic Regression for broad surveillance, XGBoost for the high-confidence investigation queue.

8. Head-to-Head: Supervised vs. Unsupervised

Comparing supervised and unsupervised on the same 1,623-provider test set, with the same features and the same seed, is the part I think is most useful. The question isn't whether labels help — of course they do. The question is how much, and what to do before they exist. Best unsupervised model: Autoencoder at ROC-AUC 0.812. Worst supervised model: neural network at 0.930. Best supervised: Logistic Regression at 0.953. So labels buy you about 0.12 to 0.14 AUC points.

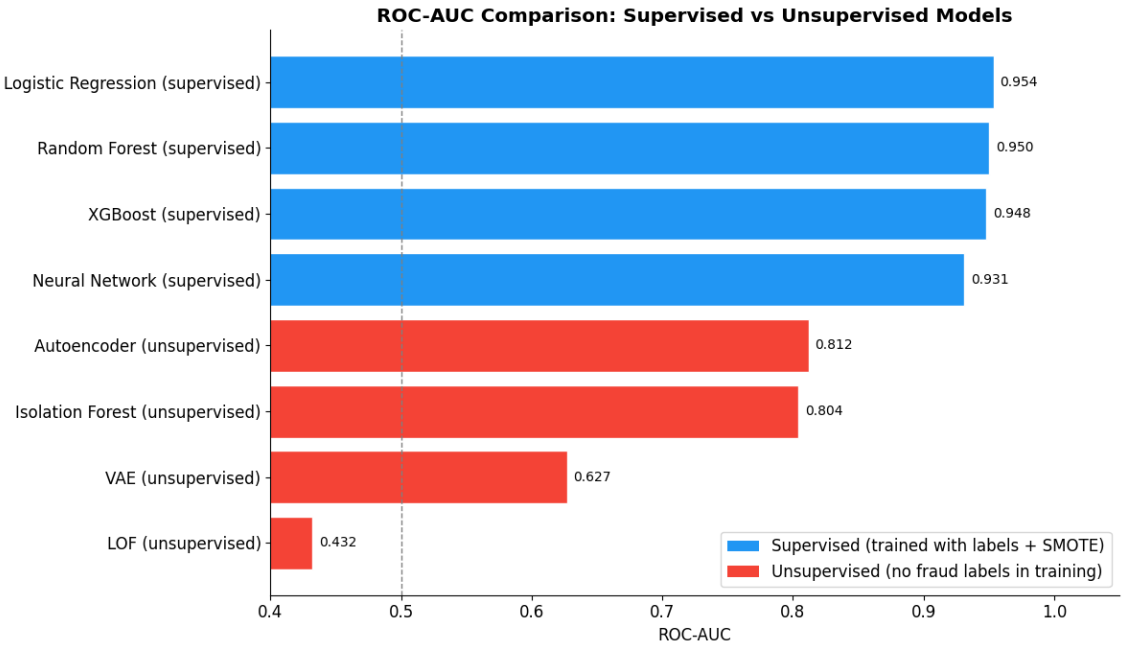

Figure 17. ROC-AUC comparison of supervised and unsupervised models. Labels raise the ceiling, but unsupervised models are useful before labels exist.

That gap argues for a deployment lifecycle, not a single winner. Stage one: no confirmed labels yet — train an Autoencoder or Isolation Forest on normal-looking provider behavior and use it to pick the first batch of investigations. Stage two: investigators confirm true and false positives, and labels start accumulating. Stage three: supervised models trained on those labels take over for higher-precision ranking. The unsupervised system doesn't retire when supervised comes online, though — it's still the right tool for catching novel patterns that aren't represented in the labeled training set.

9. Financial Impact Analysis

9.1 ROI results

The financial numbers put the AUCs in concrete terms. At a top-10% alert budget and $1,000 per false-positive investigation, the Autoencoder finds 51 true positives and 112 false positives. Those 51 providers received $68.8M in test-set reimbursement, leaving $68.7M net ROI after investigation cost. Isolation Forest comes in second at $56.0M, the VAE at $49.8M, LOF at $19.4M. Supervised models pull more dollars because they catch more fraud providers: Logistic Regression recovers $101.9M on the test set across 131 true positives; XGBoost recovers $96.0M with 106 true positives and less than half the false positives Logistic Regression carries.

Table 7. ROI at $1,000 false-positive investigation cost

Family	Model	True positives	False positives	Recoverable test set	Net ROI test set	Scaled recovery
Unsupervised	Autoencoder	51	112	$68.8M	$68.7M	~$229M
Unsupervised	Isolation Forest	40	123	$56.1M	$56.0M	~$187M
Unsupervised	VAE	31	132	$49.9M	$49.8M	~$166M
Unsupervised	LOF	11	152	$19.5M	$19.4M	~$65M
Supervised	Logistic Regression	131	143	$101.9M	$101.8M	~$340M
Supervised	Random Forest	110	87	$97.2M	$97.1M	~$324M
Supervised	XGBoost	106	60	$96.0M	$95.9M	~$320M
Supervised	Neural Network	102	64	$93.8M	$93.8M	~$313M

9.2 Sensitivity to false-positive cost

Sensitivity to false-positive cost is smaller than I expected. Recoverable reimbursement is large enough to dwarf investigation cost. Moving the Autoencoder from $500 to $2,000 per false positive shifts net ROI from $68.8M to $68.6M. Logistic Regression goes from $101.8M to $101.6M over the same range. That doesn't mean false positives don't matter — they cost investigator time, create friction with providers, and have compliance implications I'm not modeling. It means the dominant driver in this dataset is recall over the high-dollar fraudulent providers. Precision still matters when investigator capacity is the binding constraint, which is why XGBoost looks attractive even though its raw recovery is slightly lower.

9.3 Operational implications

No part of this should be read as automatic accusation. The model produces a prioritized queue; the queue gets enriched with feature-level explanations, historical trends, and claim samples; humans review. The right workflow is tiered: Autoencoder or Isolation Forest builds the cold-start queue, investigative outcomes turn into labels, supervised models train on those labels, and unsupervised scores keep running in parallel to catch what the labeled data can't represent yet. The ROI framework also makes a governance point worth taking seriously: high-dollar false negatives matter more than the usual focus on false positives, because a handful of missed high-reimbursement providers can dominate the economics in either direction.

10. Discussion

10.1 Extending the baseline

Slavov was right to be skeptical about classical methods in mixed healthcare data — LOF lands worse than random here, which is about as strong a confirmation as you can ask for. But "classical methods struggle" isn't the same as "unsupervised detection doesn't work." Isolation Forest and the Autoencoder both substantially clear LOF, and the Autoencoder closes to within roughly 0.12-0.14 ROC-AUC of the supervised ceiling without ever seeing a fraud label. With non-linear representation learning and a real held-out test design, unsupervised becomes a viable first stage, not just an exploratory pass.

10.2 Methodological lessons

A few methodological things came out of this that I want to flag. Label leakage can sneak into unsupervised projects too, if thresholds, transformations, or model choices get tuned on the full labeled data — keeping a fixed test set from the start is the cheap defense. Theoretical and operational thresholds answer different questions: the Isolation Forest score > 0.5 cutoff is useful when you have no labels at all, but top-k thresholds map more directly to real investigation capacity. And bootstrap confidence intervals turned out to matter more than I initially thought — with only 152 fraud providers in the test set, point estimates make small differences look more certain than they are.

10.3 Mixed-data evidence

The results make a fairly concrete empirical case that this kind of mixed feature space defeats local-density reasoning. "High-dimensional" isn't really the right frame — the issue is the blend of skewed dollar amounts, count variables, proportions, missingness-driven features, demographic controls, and temporal derivatives. There's no coherent notion of "neighborhood" across that mix, so LOF can't get traction. Autoencoders work because they're learning a compressed model of normal behavior, not estimating density. Isolation Forest works because it doesn't try to estimate density at all.

10.4 Limitations

A few limitations worth being honest about. The labels are PotentialFraud, not adjudicated convictions, so the financial estimates are recoverable exposure under a prioritization rule rather than guaranteed legal recovery. The data is one payer context over a fixed historical window — temporal generalization isn't tested. SMOTE can interpolate among known fraud examples but can't conjure fundamentally new fraud patterns. Provider aggregation buys interpretability but discards claim-level detail that might help explain *which* specific claims triggered suspicion. And the feature set is fixed — better encoding of diagnosis and procedure sequences would probably help both families of models, but I didn't pursue it here.

11. Conclusion

Provider-level Medicare fraud detection benefits from both unsupervised and supervised methods, but at different points in the lifecycle. The Autoencoder is the best cold-start model — ROC-AUC 0.812, PR-AUC 0.332, no fraud labels in training. Isolation Forest is nearly tied and has the practical advantage of a theory-based threshold that doesn't require knowing the fraud rate. LOF doesn't work here, which validates the original concern about classical density methods on this kind of mixed claims data. The VAE underperforms the plain Autoencoder, and I think the KL penalty is the reason — it smooths away the same reconstruction failures that signal fraud.

Labels lift the ceiling substantially. Logistic Regression ranks best on ROC-AUC and recall; XGBoost gives the best precision/F1 tradeoff. The 0.12-0.14 AUC gap from best unsupervised to best supervised is roughly what labeled investigations are worth in this setting. In dollars: ~$68.8M of recoverable reimbursement from the Autoencoder on the test set (~$229M scaled), against ~$101.9M from Logistic Regression (~$340M scaled). Not legal recoveries — prioritization upper bounds — but enough to make the case that ranking matters.

The recommendation is staged deployment. Start with unsupervised to rank providers before any confirmed labels exist. Investigate the top of the queue. Convert outcomes into labels. Train supervised models and shift to them for higher-precision operations. Keep unsupervised running in parallel for whatever the labels can't capture yet. Unsupervised solves the cold start; supervised exploits the institutional knowledge that successful investigations create.

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